

Topic 5: Chemical energetics - Answers

IGCSE Chemistry 0620 Paper 4 topical answer booklet.

Subtopic: Exothermic/endothemic changes and bond energies

Questions in this answer PDF: 4

Use this with the matching topical question PDFs. The answer points are organised by the same source labels.

Short explanation notes are added for harder calculation, electrolysis, equilibrium, rate, salts, organic, metals and practical questions.

Periodic table reference pages are not included to save printing paper.

2020 May/June Paper 43 Question 6

Main topic: Topic 5: Chemical energetics | Marks: 5 | Answer source: 0620_s20_ms_43.pdf

Extra topic(s): Topic 12: Experimental techniques and chemical analysis, Topic 2: Atoms, elements and compounds, Topic 3: Stoichiometry

Answer points

6(a)(i)

- 2, 2, 4, 1

6(a)(ii)

- thermal decomposition

6(b)

- -

- add aqueous sodium hydroxide

- -

- then (reduction with) aluminium (foil) (and warm)

- -

- (ammonia gas produced which) turns damp red litmus blue

Easy explanation / reminder

- *Practical tip: name the reagent/apparatus and give the positive observation. Observation marks need colour changes or precipitates clearly stated.*

- *Energy tip: exothermic releases energy and products are lower. Endothermic absorbs energy and products are higher.*

2021 Oct/Nov Paper 42 Question 4

Main topic: Topic 5: Chemical energetics | Marks: 17 | Answer source: 0620_w21_ms_42.pdf

Answer points

4(a)

- E of making bonds > breaking bonds
- bond making releases energy
- bond breaking requires energy

4(b)(i)

- exothermic mark
- horizontal line below energy level to R.H.S. of reactants line
- and
- labelled $\text{COCl}_2(\text{g})$ (1)
- activation E mark
- activation energy 'hump' with upward arrow labelled / activation energy (1)
- energy change mark
- one downward arrow labelled ΔH
- and
- energy change starting from E level of reactants and finishing at E level of products (1)

4(b)(ii)

- increases rate of reaction

4(c)(i)

- equilibrium shifts to right hand side (1)
- fewer moles (of gas) on right hand side (1)

4(c)(ii)

- equilibrium shifts to left hand side (1)
- (forward) reaction is exothermic (1)

4(d)

- bond energy in making bonds
- $= [(2 \times 400) + 745] = 1545 \text{ (kJ mol}^{-1}\text{)}$
- use of total E change
- $-230 = [240 + E(\text{C}\equiv\text{O})] - 1545$
- OR
- $[240 + E(\text{C}\equiv\text{O})] = -230 + 1545 = (+1315)$
- $E(\text{C}\equiv\text{O})$
- $= [-230 + 1545] - 240 = 1075 \text{ (kJ mol}^{-1}\text{)}$

4(e)

- all single bonding dot and cross pairs correct
- double C=O bond dot and cross pairs are correct
- complete diagram is correct

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*
- *Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.*

2023 Oct/Nov Paper 42 Question 5

Main topic: Topic 5: Chemical energetics | Marks: 16 | Answer source: 0620_w23_ms_42.pdf

Extra topic(s): Topic 6: Chemical reactions

Answer points

5(a)

- enthalpy change (of reaction)

5(b)(i)

- rate of reaction increases (1)
- more (CCl₄) particles per unit volume (1)
- frequency of collisions between (CCl₄ and H₂O) particles increases (1)

5(b)(ii)

- (position of) equilibrium moves to left hand side (1)
- more (gaseous) moles / molecules on right hand side (1)

5(b)(iii)

- Labels mark
- CCl₄(g) + 2H₂O(g) on reactant line
- AND
- CO₂(g) + 4HCl(g) on product line
- Activation E mark
- upward arrow labelled E_a from energy level of reactants to top of 'hump'
- Energy change mark
- one downward arrow labelled ΔH
- AND
- energy change starting from E level of reactants and finishing at E level of products

5(b)(iv)

- minimum energy (1)
- that colliding particles must have to react (1)

5(b)(v)

- (decreased by adding) catalyst

5(c)

- energy needed to break bonds in reactants
- $[(4 \rightarrow 340) + (4 \rightarrow 460)] = 3200 \text{ (kJ / mol) (1)}$
- energy released when bond in carbon dioxide form
- $2 \rightarrow 805 = 1610 \text{ (kJ / mol) (1)}$
- calculate H-Cl bond energy
- $3200 - (1610 + E(4\text{H-Cl})) = -130$
- $E(4\text{H-Cl}) = 3330 - 1610 = 1720 \text{ (kJ / mol) (1)}$
- $1720 / 4 = 430 \text{ (kJ / mol) (1)}$

Easy explanation / reminder

- *Calculation tip: start with moles = mass / Mr or moles = concentration x volume. Then use the mole ratio from the balanced equation before converting to the final answer.*

- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.

2025 Oct/Nov Paper 41 Question 4

Main topic: Topic 5: Chemical energetics | Marks: 16 | Answer source: 0620_w25_ms_41.pdf

Answer points

4(a)

- energy level of products below energy level of reactants
- label of $\text{SO}_2(\text{g}) + 4\text{HF}(\text{g})$ on product line
- upward arrow labelled E_a from energy level of reactants to the highest point of the curve
- one downward arrow labelled ΔH
- AND
- energy change starting from energy level of reactants and finishing at energy level of products

4(b)

- $= (4 \rightarrow \text{S-F}) + (4 \rightarrow \text{O-H}) = (4 \rightarrow 330) + (4 \rightarrow 460) = 3160$
- bond energy of products - $3160 = -54$
- $= 3160 - (-54) = 3214$
- $(2 \rightarrow \text{S=O}) + (4 \rightarrow \text{H-F}) = 3214$ therefore $2 \rightarrow \text{S=O} = 3214 - (4 \rightarrow 570) = 934$
- $\text{S=O} = \frac{1}{2} \times 2 = 934$ $\frac{1}{2} \times 2 = 467$

4(c)

- increased temperature:
- (position of) equilibrium moves to left hand side
- reaction is exothermic
- increased pressure:
- (position of) equilibrium moves to left hand side
- less (gaseous) moles on left hand side
- no change / catalyst does not affect position of equilibrium

4(d)

- kinetic energy of particles decreases
- frequency of collisions between particles decreases
- lower percentage/proportion/fraction of collisions / particles have energy greater than / equal to activation energy

Easy explanation / reminder

- Calculation tip: start with $\text{moles} = \text{mass} / M_r$ or $\text{moles} = \text{concentration} \times \text{volume}$. Then use the mole ratio from the balanced equation before converting to the final answer.
- Equilibrium tip: explain which reaction is favoured and why. A closed system is needed because reactants and products must not escape.